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TECH

INTERNATIONAL INFORMATION NOBEL CENTRE
OFFICIAL REPRESENTATION IN UKRAINE

LLC SOKOLOVSKIY-FORTUNA

UNERA LUXEMBOURG S.A.

Modern approach of improving bioeconomy, ecology and health.



LLC SOKOLOVSKIY-FORTUNA

ABOUT US

SPIRULINA

PROBLEM & SOLUTION

OPEN – CELL SPIRULINA

PRODUCT FOR POULTRY

WHO ABOUT SPIRULINA

SCIENTIFIC RESEARCH

Our company [LLC SOKOLOVSKIY-FORTUNA](#) together with [UNERA Luxembourg S.A.](#) we focus on R&D of innovative bio products and solutions using sustainable matter and high-tech biotechnologies to improve social and economic needs.

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Spirulina is a type of micro-algae, also known as cyanobacteria. It has a long history of use as food and it has been reported to be used during the Aztec civilization.

Spirulina Platensis is rich in proteins with a very high biological value, which make up, up to **70%** of the composition of spirulina with antioxidant and other vital properties.

In the last few years, number of studies have demonstrated that Spirulina not only has beneficial effects in pharmacological applications but also in the agricultural industry.



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SPIRULINA COMPOSITION:

The Phycobiliproteins and Phycobilins	Phycocyanins, Phycoerythrins, Allophycocyanins and other Chromophores Protein-pigments
The Phytochrome	Phytochrome red, Phytochrome far-red
Interchangeable amino acids	Histidine, Isoleucine, Leucine, Lysine, Methionine, Phenylalanine, Threonine, Tryptophan, Valine, Tyrosine, Cysteine, and others
Replaceable amino acids	Alanine, Arginine, Aspartic acid, Cystine, Glutamic acid, Glycine, Proline, Serine, Tyrosine, and other
Nucleotide	Adenine, Cytosine, Uracil, Guanine
Deoxyribonucleic acid - dsDNA	
Deoxyribonucleic acid - ssDNA	
Ribonucleic acid - RNA	
Fatty acids	Lauric, Palmitic, Palmitoleic, Heptadecanoic, Stearic, Oleic, Linoleic, Omega-3, Omega-6, Omega-9, Arachidonic, Behenic, Erucic, Lignoceric, and other
Chlorophylls	Photosynthetic natural pigments Chlorophyll-a, Chlorophyll-b, Bacterial-Chlorophyll-b (chlorin e6) and other
Carotenoids and Xanthophylls	Fat-soluble natural pigments Carotenes, Xanthophylls, Zeaxanthin, α-Carotene, β-Carotene, γ-Carotene, Lycopene and other
Vitamins	A, B1, B2, B3, B6, B12, D3, K1 and other
Minerals	Calcium, Phosphorus, Iron, Manganese, Sodium, Chlorine, Magnesium, Zinc, Potassium, Copper, Chromium, Iodine, Molybdenum, Selenium, and other
Other	Prokaryotes, Eukaryotes, Prebiotics, Polysaccharides, Carbohydrates and dietary fibers

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Spirulina is a gram-negative, microscopic, filamentous cyanobacterium that grows in alkaline water bodies. It is extensively utilized as a **nutraceutical food supplement** all over the world due to its high levels of functional compounds, such as phycocyanin, phenols and polysaccharides, with anti-inflammatory, antioxidant, immunomodulating properties both in vivo and in vitro.

PROBLEM: impossible application of SPIRULINA in commercial poultry farms and in agriculture, due to high value of proteins, organic material, size of particles and INDIGESTABILITY. Ineffective absorption of active elements due to Gram – Negative Cell Membrane.

SOLUTION: **LIQUEFIED SPIRULINA CELLS** also known as **OPEN - CELL SPIRULINA** innovative raw feed material product for poultry, applicable via water drinking system, while making all beneficial elements of SPIRULINA DIGESTIBLE and in fully accessible form for the living organism to absorb.

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TECHNOLOGY:

Applying unique and patented technology: by UNERA Luxembourg S.A. which is also known as (mild disrupted cell technology) allows to obtain a unique product

OPEN – CELL SPIRULINA.

SPIRULINA

PRESENT TECHNOLOGICAL SOLUTION ENABLES:

Complete accessibility and availability of beneficial elements of Spirulina, in biologically active unchanged chemical composition. Technological solution allows to guarantee quality of lysed biomass, preserving maximally unchanged chemical composition and biological activity of released bioproduct **due to the absence of any factors damaging or reducing such activity** (as high temperature, aggressive and environmentally hazardous chemical substances (e.g. alkalis, acids, abrasive surface-active substances, detergents)) or long-lasting high power pressure.

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ELEMENTS:

Phycocyanin, Tryptophan, Phenylalanine, Omega-3, Omega-6, Omega-9, Zeaxanthin, α -Carotene, β -Carotene, Minerals and Vitamins A, B1, B2, B3, B6, B12, D3, K1, Calcium, Iron, Selenium and many other.

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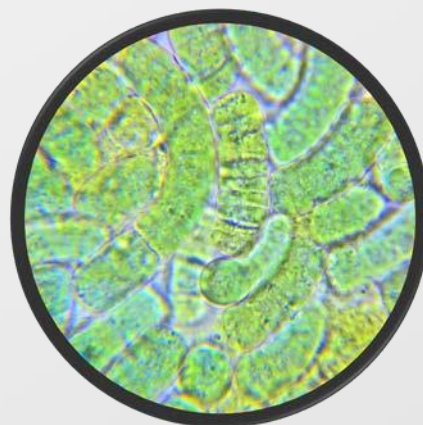
PROBLEM & SOLUTION

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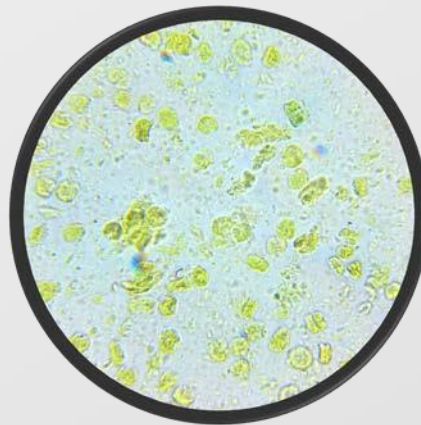
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PROCESS:



BEFORE



AFTER



RESULT



FINAL PRODUCT

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SIMPLIFIED VISUAL EXPLANATION “OPEN – CELL”:



INCREASED

1. BIOACCESSIBILITY
2. BIOAVAILABILITY

NO CHANGE IN CHEMICAL COMPOSITION!

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PROCUREMENT OF RAW MATERIAL:

SPIRULINA

United States of America, California.

Certified with highest standards.

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PRODUCT: LIQUEFIED ALGAE SPIRULINA FEED MATERIAL FOR BROILER & LAYING HENS.

SPIRULINA

DESCRIPTION: Presents new concept in the market as liquefied bioactive sustainable feed product, with natural enhancing properties for more cost-effective growth.

PROBLEM & SOLUTION

Problem solving and easy applicable product, specially formulated for use in commercial poultry farms.

OPEN – CELL SPIRULINA

It can be seen more as supplement (complementing already established diet) which enhances more cost-effective growth and welfare for poultry, by acting as enzyme, optimizing nutrient digestibility and performance, with functional properties, such as antioxidant, anti-inflammatory, anticarcinogenic and antimicrobial effects.

WHO ABOUT SPIRULINA

Recent study by research institution (Vilnius University) identified that our product has antibacterial properties including **against Campylobacter**.

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APPEARANCE: Liquid.

Created by applying unique patented technology from **LUXEMBOURG***

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RECORDED BENEFITS:

↑ HIGHER:

- Growth Coefficient
- Egg Quality & Quantity & Weight
- Meat Quality
- Potential Total Yield
- Health and Immune System
- Cost Savings
- Welfare of animals

↓ LOWERED:

- Mortality Rate
- FCR (Feed Conversion Ratio) / More effective consumption of feed*
- Pollution of underground waters
- Use of Antibiotics
- Risk of illness

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EASY APPLICATION IN POULTRY FARMS:

- DIRECTLY in poultry house / building.
- Via WATER drinking system.
- Using "DOSATRON" or similar system.

01.

- MINIMUM one bottle per day for one poultry house / building.
- During day time.
- Application length per day, from 6 to 12 hours.

02.

- Application duration: Throughout all life cycle of poultry.

03.

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ADDITIONAL COMMERCIAL BENEFITS:

COST SAVING 01.	HIGHER POULTRY WELFARE 02.	LOWER CARBON FOOTPRINT 03.
NEW MARKETING OPPORTUNITIES 03.	IMPROVED PERCEPTION & AWARENESS OF BRAND 04.	LITHUANIA LUXEMBOURG USA CO-OPERATION 05.

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WORLD HEALTH ORGANISATION:



World Health
Organization

The WORLD HEALTH ORGANISATION has described Spirulina as one of the **greatest super foods** on earth and NASA considers it as an excellent compact food for space travel, as small amount can provide a wide range of nutrients.

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SPIRULINA FOR POULTRY:

- Spirulina Platensis in Poultry Nutrition

<https://www.cambridgescholars.com/resources/pdfs/978-1-5275-2128-5-sample.pdf>

- Digestive Constraints of Arthrospira platensis in Poultry and Swine Feeding

<https://pubmed.ncbi.nlm.nih.gov/36230060/>

- Modulating the Growth, in Broiler Chickens by Adding Dietary Spirulina Platensis Phycocyanin

<https://pubmed.ncbi.nlm.nih.gov/35624855/>

- Comparison of marine algae (Spirulina platensis) and synthetic pigment in enhancing egg yolk colour of laying hens

https://www.researchgate.net/publication/51747660_Comparison_of_marine_algae_Spirulina_platensis_and_synthetic_pigment_in_enhancing_egg_yolk_colour_of_laying_hens

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Let's focus together on life changing innovation.

Thank you for your attention.